Design and Implementation of a Line Following and Obstacle Avoidance Robot

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***Abstract*—This paper presents a low-cost line-following and obstacle-avoidance robot built on an Arduino Uno. Two IR sensors enable line tracking, while two HC-SR04 ultrasonic sensors handle obstacle detection; an L293D motor driver controls the two DC motors. Prior to construction, a systems engineering approach using SysML diagrams (block definition, internal block, activity, sequence, use case, and state machine) modeled the robot's structure and behavior. The design follows a track, avoids the first detected obstacle, then performs a 180° turn at a second obstacle before retracing the line in reverse. Testing on a custom track with curves and obstacles yielded a 100 percent success rate across 10 trials, confirming reliable performance.**

***Index Terms*—Line follower robot, IR sensor, Ultrasonic sensor, Arduino, Obstacle avoidance, DC motor**

I. Introduction and Objective

A. Introduction

Automated Guided Vehicles (AGVs) are smart, driverless vehicles that bring the speed and precision of modern automation to material handling. By using advanced embedded systems to navigate safely and adapt in real time, they turn manual logistics into a predictable, data-driven backbone for Industry 4.0 [1, 2].

Line following and collision avoidance serve as the fundamental navigation and safety pillars. Together, these embedded subsystems ensure the AGV can optimize space by working seamlessly alongside human workers while strictly complying with industrial safety standards like ISO 3691-4.

This project focuses on the systems engineering, prototyping, and development of a compact, microcontroller-driven autonomous vehicle designed for precise trajectory tracking and proactive collision mitigation.

B. Objective

The primary goal of this project is to implement, test, and validate a working prototype of an autonomous mobile vehicle using model-based prototyping principles. To meet the course milestones, the system must achieve the following specific technical objectives:

- Line Following: Use a set of infrared (IR) sensors and standard if-else logic to control the motors so the car cleanly follows a track line.
- Obstacle Detection and Avoidance: Integrate a pair of ultrasonic sensors to monitor front proximity metrics and implement a maneuver sequence to safely bypass detected obstructions before reacquiring the primary track line.
- Secondary Obstacle Detection and Turnaround: Program the car to use its front ultrasonic sensors to detect a second obstacle along the path, stop moving forward, spin 180 degrees on the spot, and follow the track back in the opposite direction.
- Power and Motor Integration: Wire the hardware so that a motor controller takes a 12V battery source, drops it down to 5V to safely power the Arduino, and handles the high-current demands of the DC motors.

II. System Design and Engineering Approach

A. System Overview

The Line Following Robot is an autonomous system designed to follow a predefined path while detecting obstacles during operation. The system combines sensing, control, and actuation components to achieve reliable navigation. A systems engineering approach was applied throughout the project to identify requirements, define subsystem interactions, and develop an organized system architecture before implementation.

The robot consists of four main subsystems: the sensing subsystem, control subsystem, actuation subsystem, and power subsystem. These subsystems work together to support autonomous movement and obstacle detection.

B. System Architecture

A Block Definition Diagram (BDD) was created to represent the main components of the robot and their relationships. The diagram provides an overview of how the sensing, control, movement, and power elements are organized within the system.

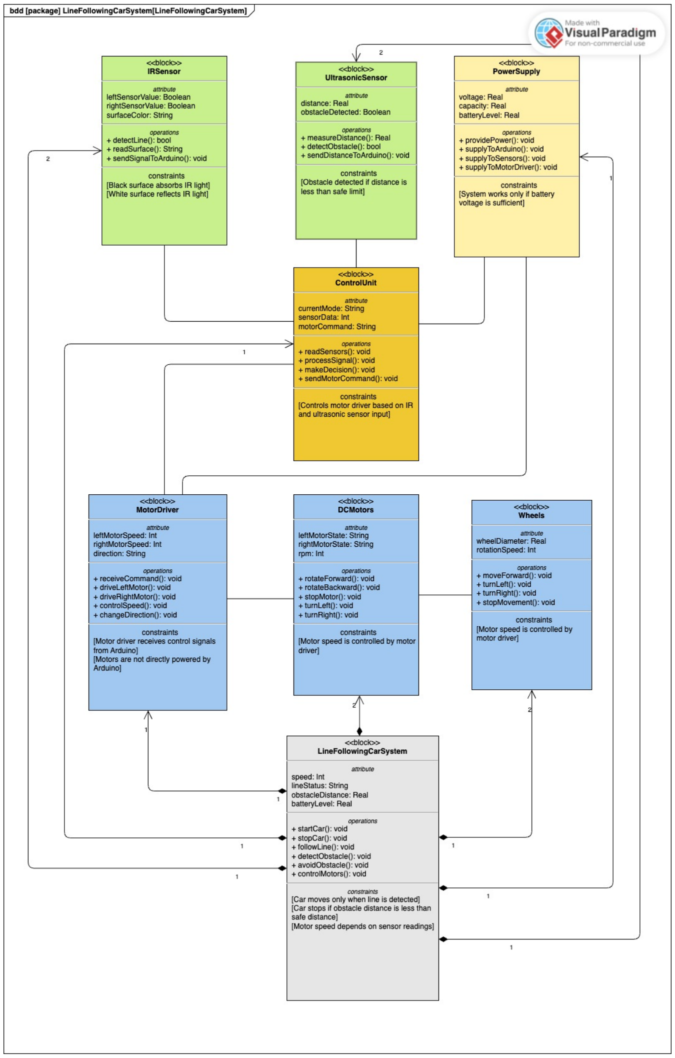

Fig. 1. Block Definition Diagram of the Line Following Robot

The sensing subsystem includes infrared (IR) sensors for line detection and ultrasonic sensors for obstacle detection. The control subsystem is based on the Arduino Uno microcontroller, which processes sensor information and determines the required motor actions. The actuation subsystem consists of the L293D motor driver and two DC motors that provide movement. The power subsystem supplies electrical energy to all hardware components.

C. Internal Component Interaction

An Internal Block Diagram (IBD) was developed to illustrate the connections and interactions between system components. The diagram shows how information and control signals are exchanged within the robot.

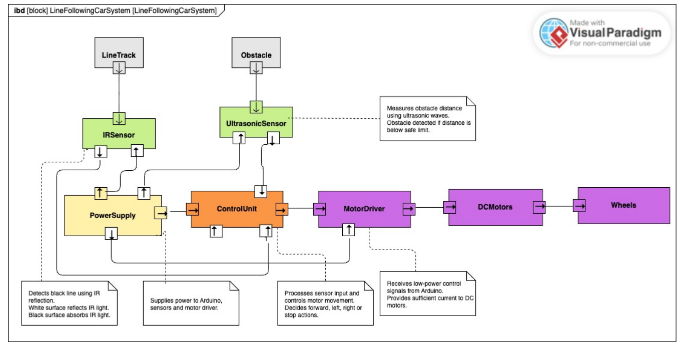

Fig. 2. Internal Block Diagram

The IR sensors continuously monitor the line position and send data to the Arduino controller. At the same time, the ultrasonic sensors measure the distance to nearby obstacles. Based on these inputs, the Arduino generates appropriate control signals for the motor driver, which adjusts the speed and direction of the motors.

D. Functional Behaviour

The behaviour of the robot was analysed using system modelling techniques. An Activity Diagram was created to describe the overall operational workflow.

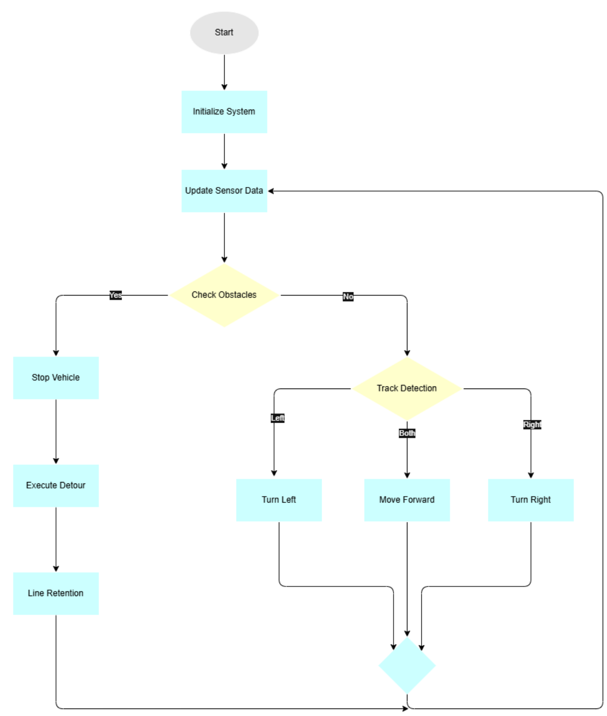

Fig. 3. Activity Diagram

The process begins with system initialization. The robot then continuously reads sensor data, evaluates the position of the line and the presence of obstacles, determines the required action, and updates motor commands. This cycle is repeated throughout operation.

A Sequence Diagram was also developed to represent the interaction between system components during robot operation.

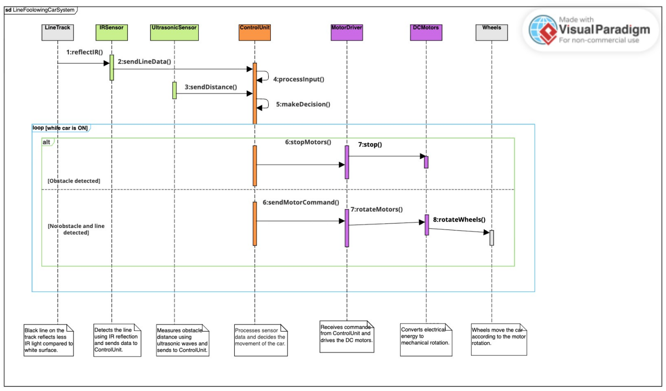

Fig. 4. Sequence Diagram

The sequence diagram illustrates how information is exchanged between the sensors, controller, and actuators during line-following and obstacle-detection activities.

E. Use Case Analysis

A Use Case Diagram was developed to identify the main functions of the robot from the user's perspective.

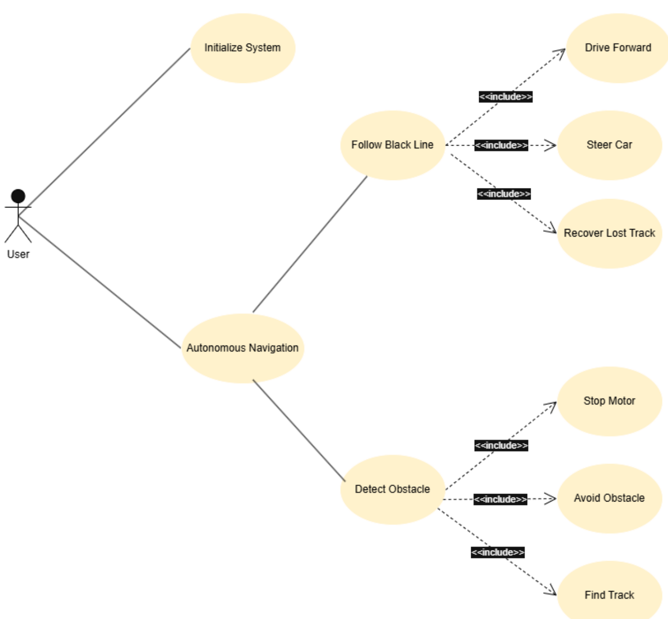

Fig. 5. Use Case Diagram

The main use cases include starting the robot, following the line, detecting obstacles, controlling movement, and stopping the robot. These functions represent the primary operational requirements of the system.

F. System States

A State Machine Diagram was created to model the different operational states of the robot.

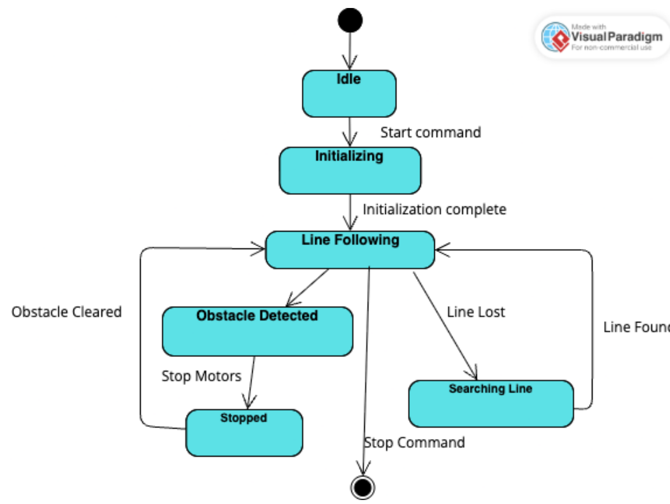

Fig. 6. State Machine Diagram

The robot initially remains in the Idle state. Once activated, it transitions to the Line Following state. When an obstacle is detected, the system moves to the Obstacle Detected state and performs a stop or avoidance action. After the obstacle is cleared, the robot returns to normal operation.

G. Design Decisions

Several design decisions were made during the development of the prototype. The Arduino Uno was selected because it is easy to program, widely supported, and compatible with various sensors and motor drivers. IR sensors were chosen for accurate line detection, while ultrasonic sensors were used to provide reliable obstacle detection. The L293D motor driver was selected because it allows simple and effective control of two DC motors.

The use of SysML diagrams improved system understanding by providing both structural and behavioural views of the robot. These models supported the implementation process and helped ensure consistency between system requirements and the final prototype design.

III. Hardware Software Implementation

A. Hardware Implementation

The developed prototype consists of a compact autonomous mobile robot capable of line following and obstacle avoidance. The system integrates sensing, control, actuation, and power management into a single embedded platform. Figure 7 shows the completed robot prototype used during implementation and testing.

The Arduino Uno microcontroller serves as the central controller of the robot. It continuously receives data from two infrared (IR) sensors and two HC-SR04 ultrasonic sensors, processes the sensor information, and generates motor control commands. The IR sensors are mounted at the front of the robot to detect the black line, while the ultrasonic sensors continuously monitor the distance to obstacles during navigation.

An L293D motor driver is used to control two DC geared motors. The motor driver receives PWM signals from the Arduino and regulates the speed and direction of both motors. To compensate for mechanical differences between the motors, calibrated PWM values were applied to obtain stable straight-line movement.

The complete system is powered by a 12 V battery. The motor driver supplies a regulated 5 V output to power the Arduino Uno and the connected sensors. This configuration provides a simple and reliable power distribution while maintaining stable operation of both the electronic and mechanical components.

The electronic components are mounted on a lightweight wooden chassis using a breadboard for rapid prototyping. This modular arrangement simplifies wiring, maintenance, and future hardware modifications.

TABLE I
HARDWARE COMPONENTS

Component	Qty	Purpose
Arduino Uno	1	Main controller
L293D Motor Driver	1	Motor control
DC Motors	2	Robot movement
IR Sensors	2	Line detection
HC-SR04 Sensors	2	Obstacle detection
Breadboard	1	Circuit prototyping
12 V Battery	1	Power supply
Wooden Chassis	1	Mechanical support

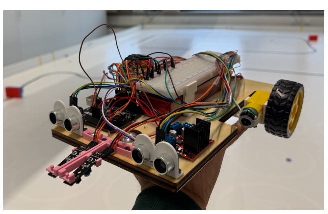

Fig. 7. Final hardware prototype of the autonomous line-following and obstacle-avoidance robot.

B. Software Implementation

The control software was developed using the Arduino Integrated Development Environment (Arduino IDE) in C++. A modular programming approach was adopted to separate sensor acquisition, motor control, obstacle detection, and navigation into independent software functions. This improves readability, simplifies debugging, and allows future system extensions.

During initialization, the Arduino configures all digital input and output pins, initializes the infrared and ultrasonic sensors, and enables the motor driver. After initialization, the software repeatedly executes the main control loop throughout the robot's operation.

The first stage of the control algorithm continuously checks the ultrasonic sensors for nearby obstacles. When an obstacle is detected within the predefined safety distance, the robot immediately stops and confirms the measurement before executing the obstacle avoidance routine.

If no obstacle is detected, the robot performs line following using the two infrared sensors. Based on the sensor readings, the controller determines whether the robot should move forward, turn left, or turn right. Independent PWM control of the two motors enables smooth steering corrections while following the track.

To improve navigation reliability, the software stores the previous movement direction. If both infrared sensors temporarily lose the line, the robot searches in the last known direction until the track is detected again. After avoiding an obstacle, the robot continues searching until the line is reacquired before resuming normal navigation.

Overall, the implemented software provides reliable autonomous operation while maintaining a computationally efficient control strategy suitable for embedded systems.

C. Hardware–Software Integration

The hardware and software operate together as a closed-loop embedded system. Sensor information collected by the infrared and ultrasonic sensors is processed by the Arduino Uno, which determines the appropriate movement commands. These commands are transmitted to the L293D motor driver, which controls the speed and direction of the two DC motors.

The continuous interaction between sensing, processing, and actuation enables the robot to follow a predefined path, detect obstacles, execute avoidance manoeuvres, and return to the original track without external intervention. The implementation demonstrates that reliable autonomous behaviour can be achieved using low-cost hardware combined with an efficient embedded control algorithm.

IV. SIMULATION AND TESTING RESULTS

A. Experimental Setup

After completing the hardware and software implementation, the robot was evaluated on a custom test track consisting of a 2.5 cm wide black line. The track contained both gentle and sharp curves to evaluate the line-following performance under different path conditions. Two obstacles were also positioned along the course to test the obstacle detection, avoidance, and turnaround functions.

The robot was tested repeatedly under similar operating conditions. During each trial, it was required to follow the black line autonomously, detect the first obstacle, execute the avoidance manoeuvre, reacquire the line, detect the second obstacle, perform a 180° turn, and continue navigation without external control.

B. Experimental Results

Multiple experimental trials were conducted to evaluate the overall performance of the system. The robot successfully followed the predefined line through both gentle and sharp curves. The infrared sensors continuously detected the position of the track and allowed the controller to make the required steering corrections.

When the first obstacle was detected, the ultrasonic sensors triggered the obstacle-avoidance routine. The robot successfully moved around the obstacle, searched for the black line, and resumed normal line-following operation after reacquiring the track.

After passing the first obstacle, the robot continued along the course until the second obstacle was detected. At this stage, the controller executed the programmed 180° turning manoeuvre. The robot then followed the line in the opposite direction as required by the test sequence.

The system was tested many times, and ten successful complete trials were recorded. During these trials, the robot completed the intended navigation sequence without manual intervention, corresponding to a recorded success rate of 100%.

TABLE II
EXPERIMENTAL PERFORMANCE SUMMARY

Test Item	**Result**
Line following	Successful
Gentle curves	Successful
Sharp curves	Successful
First obstacle detection	Successful
Obstacle avoidance	Successful
Line reacquisition	Successful
Second obstacle detection	Successful
180° turning manoeuvre	Successful
Complete successful trials	10 out of 10
Overall success rate	100%

C. Performance Evaluation

The experimental results show that the developed robot achieved the main project objectives. The combination of infrared line sensing, ultrasonic obstacle detection, and embedded motor control enabled the robot to navigate the complete test course successfully.

The robot moved at a sufficient operating speed while maintaining stable line-following performance. It was able to respond correctly to both obstacles and recover the track after the avoidance manoeuvre. The successful completion of ten recorded trials indicates that the implemented hardware and software provided reliable autonomous behaviour under the tested conditions.

V. CHALLENGES AND SOLUTIONS

During the engineering and prototyping phases, several technical challenges were encountered across the hardware and software subsystems. These obstacles were systematically resolved through testing, software adjustments, and control refinements.

A. Motor Speed Inconsistency and Straight-Line Drift A primary physical challenge involved mechanical and electrical variances between the two DC motors. When driven with identical pulse-width modulation (PWM) signals, differences in motor resistance and internal gear friction caused the robot to drift off course rather than moving straight.

Solution: To achieve balanced linear movement, a software calibration procedure was performed using trial and error. By iteratively tuning and assigning individual offset PWM values to each motor, the rotational speeds were balanced to ensure straight-line movement.

B. Track Detection Strategy During line-following testing, initial attempts were made to navigate by tracking the white boundary outline surrounding the black track. However, this approach resulted in inconsistent sensor readings. Testing confirmed that tracking the solid black line directly produced significantly better results.

Solution: The control logic and tracking strategy were reconfigured to target the solid black line directly. This adjustment provided higher contrast for the infrared sensors, leading to more reliable line tracking and smoother steering responses.

C. Ultrasonic Sensor Noise and Ghost Obstacles During proximity measurement, the HC-SR04 ultrasonic sensors occasionally received false echo reflections from surrounding walls and non-target surfaces. This noise created "ghost obstacles," resulting in false-positive detections that caused the robot to stop prematurely.

Solution: Software-level signal filtering was integrated into the obstacle-detection routine. The control algorithm was programmed to prouduce 2 consecutive range readings before triggering an action, ensuring avoidance maneuvers were only executed when a real obstacle was confirmed.

VI. CONCLUSION AND FUTURE IMPROVEMENTS

A. Conclusion

This project successfully demonstrated the design and development of a Line Following Robot capable of autonomous navigation and obstacle detection. By integrating infrared sensors, ultrasonic sensors, an Arduino Uno microcontroller, and a motor control system, the robot was able to follow a predefined path while responding to environmental conditions.

A systems engineering approach was applied throughout the project to support requirement identification, system modelling, implementation, and testing. The use of SysML diagrams provided a clear representation of both the structural and behavioural aspects of the system, helping to improve understanding of component interactions and overall system functionality.

The testing results showed that the robot was able to perform the intended line-following and obstacle-detection tasks under normal operating conditions. Therefore, the project achieved its primary objective of developing a functional robotic prototype while applying key systems engineering principles.

B. Future Improvements

Although the prototype met the project requirements, several improvements could be implemented in future developments. One possible enhancement is the integration of a PID control algorithm to improve movement accuracy and provide smoother line-following performance. Additional infrared sensors could also be incorporated to increase detection reliability, particularly when navigating sharp curves or complex paths.

The mechanical design could be further optimized by improving chassis stability, wheel alignment, and component placement. These modifications may contribute to better overall performance and durability of the robot.

Furthermore, wireless communication technologies such as Bluetooth or Wi-Fi could be added to enable remote monitoring and control. Future versions may also incorporate camera-based vision systems or machine learning techniques to support more advanced navigation and obstacle-avoidance capabilities.

Overall, the proposed improvements would increase the robot's accuracy, adaptability, and functionality, making it more suitable for complex autonomous applications.

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